Technical Supplement & Parameter Index: Multi-Scale Chameleon Scalar Fields and Anisotropic Speed of Light Profiles

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Abstract

This technical supplement details the physical parameters, field equations, and boundary conditions evaluated across the 10-module integration pipeline and its associated numerical simulations. This index documents the parameter configurations utilized in the accompanying preprint repository, which evaluates a six-dimensional semiclassical action functional over an extended configuration manifold to address the local cosmological expansion discrepancy via a transient early dark energy contribution while satisfying local terrestrial screening constraints.

1 Pipeline Framework Structure

The technical supplement and its corresponding repository code operate via a 10-module synchronized calculation pipeline, structured as follows:

1. **Vacuum Baseline Array:** Initializes standard cosmological parameters and energy benchmarks.
2. **Sound Horizon Root-Solver:** Automatically derives the geometric Early Dark Energy tracking scale factor.
3. **Torsion Scale Converter:** Maps high-energy cutoff thresholds down to absolute length dimensions.
4. **Velocity Dispersion Filter:** Linearizes metric perturbations to evaluate deep-space wave propagation.
5. **Chameleon Screening Module:** Pins local scalar gradients within massive material boundaries.
6. **Lapse Decoherence Tracking:** Monitors temporal tracking phase variations under strong-field boundaries.
7. **Quantum Damping Profile:** Regulates non-unitary state suppression as the gravitational lapse vanishes.
8. **Cosmological Delay Integral:** Evaluates high-redshift multi-messenger lookback travel times.
9. **Substrate Shielding Engine:** Simulates phonon collision minimization in condensed matter.
10. **SQUID Transport Mapping:** Transforms phase-slip flux directly into readable voltage modulation curves.

2 Simulation Constants and Boundary Baselines

The numerical simulation environment fixes its baseline variables using the cosmological and environmental constraints detailed in Table 1:

Parameter Anchor	**Sym.**	**Calibrated Value**	**Physical Definition / Boundary**
Absolute vacuum speed of light	c	299792458.0 m/s	Fixed laboratory baseline
Early Dark Energy Density Peak	f_{EDE}	**0.0589**	Peak fraction at critical redshift $z_c \approx 3500$
Sound Horizon Compression Factor	Δr_s^*	**9.9497%**	Comoving sound horizon scale modification
Early Reconciled Expansion Anchor	$H_{0,\text{rec}}$	**67.95 km/s/Mpc**	Reconciled CMB acoustic horizon baseline
Maximum local velocity rift	$\Delta c_{\max}$	1.7097 m/s	Unscreened deep-space anisotropy peak
Minimum anisotropic velocity bound	$c_{\min}$	299792456.2903 m/s	Axis parameter under maximal deceleration
Effective field theory cutoff scale	Λ_{EFT}	1.0000×10^{-11} J	Short-range ultraviolet energy regulator
Reconciled Local Hubble constant	H_0	**74.39 km/s/Mpc**	Fully resolved local expansion rate

Table 1: Reconciled simulation constraints matching repository software diagnostic outputs.

3 Field Equations and Local Screening Constraints

The directional velocity modification profile is mapped relative to an external scalar gradient via the spatial metric perturbation tensor projection:

$$c_{\text{eff}}(\hat{\mathbf{k}}) = c\left[1 - \frac{\alpha_0^2 \lambda_{\text{torsion}}^2 (1-\Phi^2)^2}{2} |\nabla\Phi \times \hat{\mathbf{k}}|^2\right] \tag{1}$$

In high-density material environments ($\rho_b \gg 1.0$), the effective mass of the scalar field increases dynamically, pinning the tracking field variable to its boundary limits ($\Phi \to \pm 1$). This suppresses spatial gradients ($\nabla\Phi \to 0$), satisfying local experimental constraints down to the 10^{-17} level within cryogenic cavities to mask macroscopic vacuum anisotropies:

$$\mathcal{F}_{\text{screening}}(\rho_b) \propto e^{-\alpha_0^2 \cdot 550} \Longrightarrow c_{\text{eff}}(\theta) \longrightarrow c \tag{2}$$

Under high space-time curvature boundaries near the gravitational lapse limit ($N \to 0$), the temporal component of the field's kinetic tracking term undergoes a rapid phase divergence to preserve high-energy unitarity, causing instantaneous state decoherence:

$$\Gamma_{nn'}(N) = \frac{\alpha_0^2 \omega^2 \tau_J^2}{3N^2} \left(\frac{\hbar\omega}{\Lambda_{\text{EFT}}}\right)^3 \left(\frac{\partial\Phi}{\partial t}\right)^2 \Delta n^2 \longrightarrow +\infty \tag{3}$$

where $\tau_J \equiv \hbar/(2eI_0R)$ is the characteristic Josephson relaxation time, ω is the operational transition frequency, and Δn represents the quantum state difference. This positive divergence enforces stable non-unitary damping as the localized metric configurations near the horizon limit.

4 Cosmological Timing Delays and Coherence Shielding

Integrating photon lookback travel times across high-redshift boundaries ($z_{\text{source}} = 2.0$) relative to a gravitational wave counterpart yields an explicit arrival time separation at Earth due to the

underlying field coupling variations. Direct analytical separation isolation yields:

$$\Delta t_{\text{arrival}} = \int_0^{z_{\text{source}}} \frac{dz}{(1+z)H(z)} \left[\frac{1}{v_{\text{photon}}(z)} - \frac{1}{v_{\text{gw}}(z)} \right] = \mathbf{3116.5147}\ \text{s} \tag{4}$$

In condensed matter configurations, substrate density engineering can utilize this suppressed noise floor to isolate processing states from ambient thermal noise at room temperature (293.15 K). Under edge-reflected packet filter configurations, localized stochastic pressure gradients vanish cleanly ($\max c_s^2 = 9.2738 \times 10^{-10}$), minimizing environmental phonon collisions and preserving a local coherence fidelity plateau of approximately 78.97%.

References

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